

Loop, Mesh, Node and Branch Explained

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What is Branch?

A branch represents the single circuit elements like resistor, capacitor, inductor, voltage, or current source.

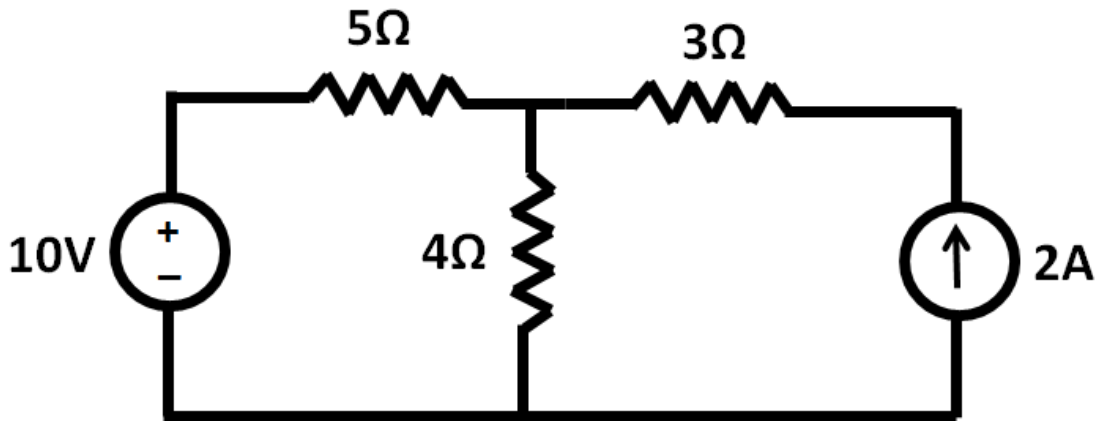


Figure. 1

For example, for the circuit shown in figure 1, there are five branches. A 10 V voltage source, 2A current source, 4 Ω , 5 Ω , and 3 Ω resistors.

What is Node?

A node is a point in the circuit where two or more circuit elements (or branches) are connected.

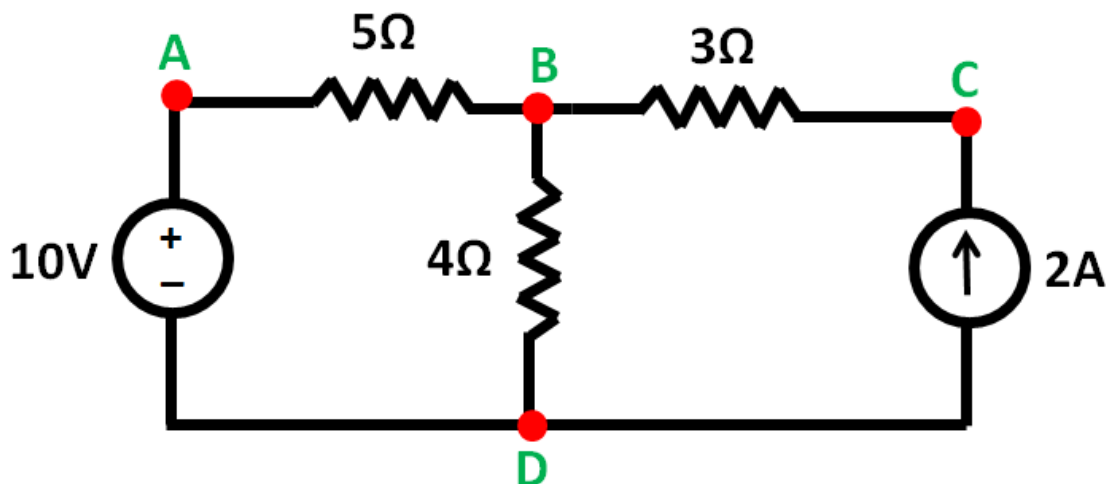


Figure. 2 (Representation of Four Nodes in the Circuit)

For example, as shown in Figure 2, the above circuit contains the Four nodes. The node A, B, C, and D.

What is Loop?

Any closed path in the circuit is called as a loop.

A loop is a closed path formed by starting at a node, passing through a set of nodes, and returning to the starting node without passing through any node more than once.

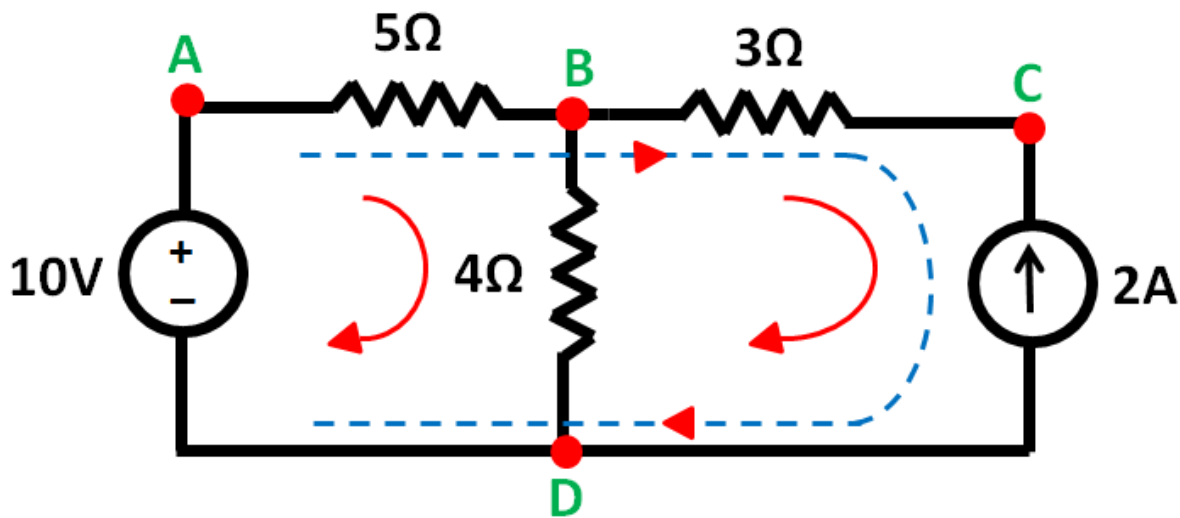


Figure. 3 (Representation of Loop in the Circuit)

For example, as shown in Figure.3, the circuit contains three loops.

The first is loop A-B-D-A, the second loop is B-C-D-B. And the third loop is A-B-C-D-A.

What is Mesh?

A mesh is a closed path in the circuit, which does not contain any other closed path inside it.

For example, as shown in Figure.3, loop 1(A-B-D-A) and loop 2 (B-C-D-B) does not contain any other closed path within them. And they are the example of the Mesh. While loop 3 (A-B-C-D-A) contains loop 1 and loop 2 within it. So, it can't be called as a Mesh.

Note: All Mesh are loops but not all the loops are Mesh.